Additional details : Linear Regression

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Outline

- Fitting non linear models using Linear Regression framework
- Robust fitting and Regularization
- Bias-variance tradeoff

Non-linear models using Linear Regression framework

- **In Linear regression can we only seek a model of the form $y = \beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n$?**

No. In linear regression, the function

- Needs to be a linear combination of the parameters $\beta_0 \;\; \beta_1 \;\; \ldots \beta_n$

- But need not necessarily have a linear relationship with respect to the independent variables ($\mathbf{x_1, x_2, \ldots, x_n}$)

A general framework for linear regression

Consider for simplicity one predictor variable and N data points.

- Fitting a linear regression model can be expressed in matrix vector form by an **overdetermined system of equations** of the form :

$$\mathbf{A\,p = b}$$

- p is a vector of unknown parameters being solved for.

- Here "=" does not indicate equality as such. Interpret **Ap** as LHS and **b** as RHS.

Predictor variable (x)	**Response variable (y)**
x_1	y_1
x_2	y_2
...	...
x_N	y_N

A general framework : Fitting a linear model

Suppose we wish to fit a model of the form $\mathbf{y = \beta_0 + \beta_1 x}$
Then :

$$A = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_N \end{bmatrix} \quad p = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} \quad b = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}$$

A general framework : Fitting a quadratic model

Suppose we wish to fit a model of the form $\mathbf{y = \beta_0 + \beta_1 x + \beta_2 x^2}$
Then :

$$A = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_N & x_N^2 \end{bmatrix} \quad p = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} \quad b = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}$$

Exercise : Write down the matrices and vectors (A,p,b) if seeking a model of the form $\mathbf{y = \beta_0 + \beta_1 x_1 + \beta_2 (\log(x_1))}$

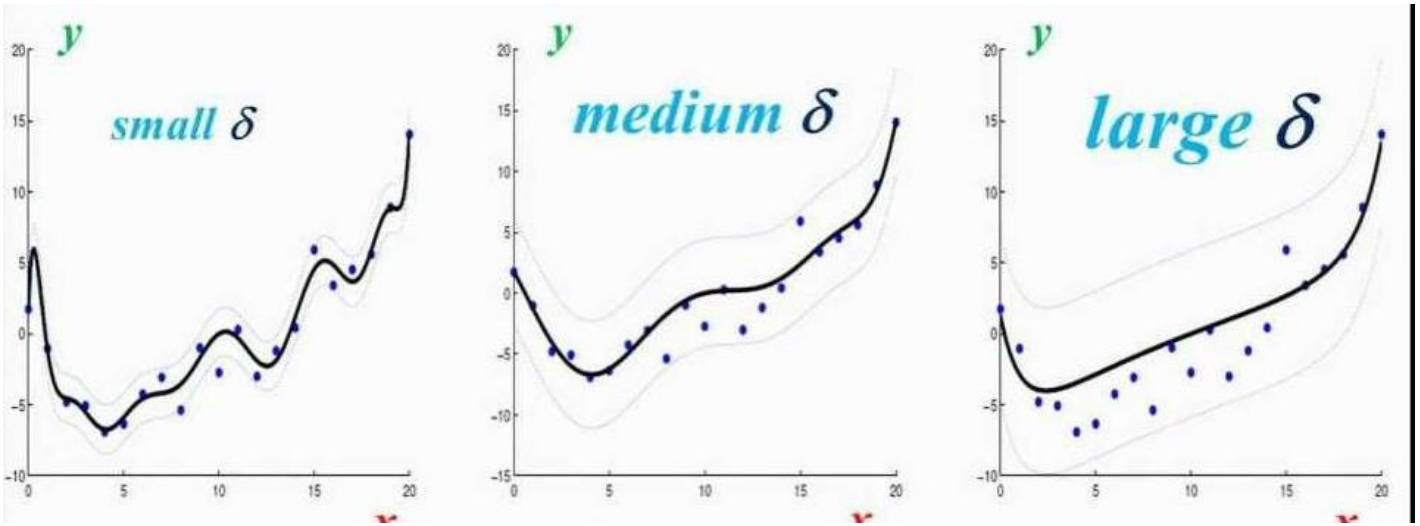
Regression with regularization

Regularization

- A process of introducing additional information in order to solve an ill-posed problem or to prevent overfitting. Regularization allows complex models to be trained on data sets of limited size without severe over-fitting, essentially by limiting the effective model complexity.

- In machine learning regularization is achieved by introducing additional terms which indirectly penalize an overly complex model (and hence mitigate overfitting).

A curve fitting example



(a)
Overfitting

(b)
Just right

(c)
Underfitting

Without regularization, fitting a high degree polynomial results in a very jagged curve (a)

Regularization results in a smoother curve while still providing a good fit to the data.(b)

Over regularization results in overly smooth curve at the cost of poor fitting (c)

Image credit : UBC Computer Science, Nando De Freitas

Recall :

In Ordinary Least Squares (OLS), we have an **overdetermined** system **Ap = b**

In Ordinary Least Squares (OLS) method the solution p^* is one which minimizes the sum of squared errors (SSE). In matrix vector notation,

$$SSE = || Ap - b ||^2 = (Ap - b)^T (Ap - b)$$

It can be shown that the Least Squares solution to the overdetermined system of equations **Ap = b** is given by the solution to the system of equations given by (also called the normal form)

$$A^T Ap = A^T b$$

The least squares solution is given by :

$$p^* = (A^T A)^{-1} A^T b$$

Regression with regularization : Ridge regression

- In Ridge regression, an additional constraint on the coefficients is imposed, via an additional penalty term
- The resultant optimization objective is a penalized residual sum of squares as given by

$$PRSS(p) = \underbrace{(b - Ap)^T (b - Ap)}_{\text{Data term}} + \underbrace{\lambda \mid\mid p \mid\mid_2^2}_{\text{Regularization term}}$$

It can be shown that the solution is : $p_\lambda^{ridge} = (A^T A + \lambda I)^{-1} A^T b$

Least absolute shrinkage and selection operator (LASSO)

- In LASSO, there is a penalty term based on L_1 norm,
  The objective function is :

$$PRSS(p) = \underbrace{(b - Ap)^T (b - Ap)}_{\textbf{Data term}} + \underbrace{\lambda \mid\mid p \mid\mid_1}_{\textbf{Regularization term}}$$

Comparison of ridge and LASSO regression

- In both ridge regression and LASSO, λ controls the influence of the regularization relative to the data term.
- In ridge regression the penalty term is an L_2 norm whereas in LASSO, the penalty term is an L_1 norm.
- The ridge regression typically produces smaller coefficients, the errors are "distributed" across the coefficients.
- The LASSO objective encourages sparsity in the model (typically a few coefficients are 0), thus indirectly performing a feature selection.

Elastic net

A convex combination of LASSO and ridge regression penalty terms. The objective function is

$$PRSS(p) = (b - Ap)^T (b - Ap) + \lambda_2 \mid\mid p \mid\mid_2^2 + \lambda_1 \mid\mid p \mid\mid_1$$

λ_1 and λ_2 control the relative influence of the L_1 and L_2 penalty terms respectively.

Bias variance trade off

Bias variance tradeoff

Consider a regression problem and the following decomposition of the expected squared loss

expected loss = $(\text{bias})^2$ + variance + noise

where

$$
\begin{aligned}
(\text{bias})^2 &= \int \{\mathbb{E}_{\mathcal{D}}[y(\mathbf{x};\mathcal{D})] - h(\mathbf{x})\}^2 p(\mathbf{x})\,\mathrm{d}\mathbf{x} \\
\text{variance} &= \int \mathbb{E}_{\mathcal{D}}\left[\{y(\mathbf{x};\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[y(\mathbf{x};\mathcal{D})]\}^2\right] p(\mathbf{x})\,\mathrm{d}\mathbf{x} \\
\text{noise} &= \int \{h(\mathbf{x}) - t\}^2 p(\mathbf{x}, t)\,\mathrm{d}\mathbf{x}\,\mathrm{d}t
\end{aligned}
$$

Bias variance trade-off

There is a trade-off between bias and variance,

- very flexible models have low bias and high variance,

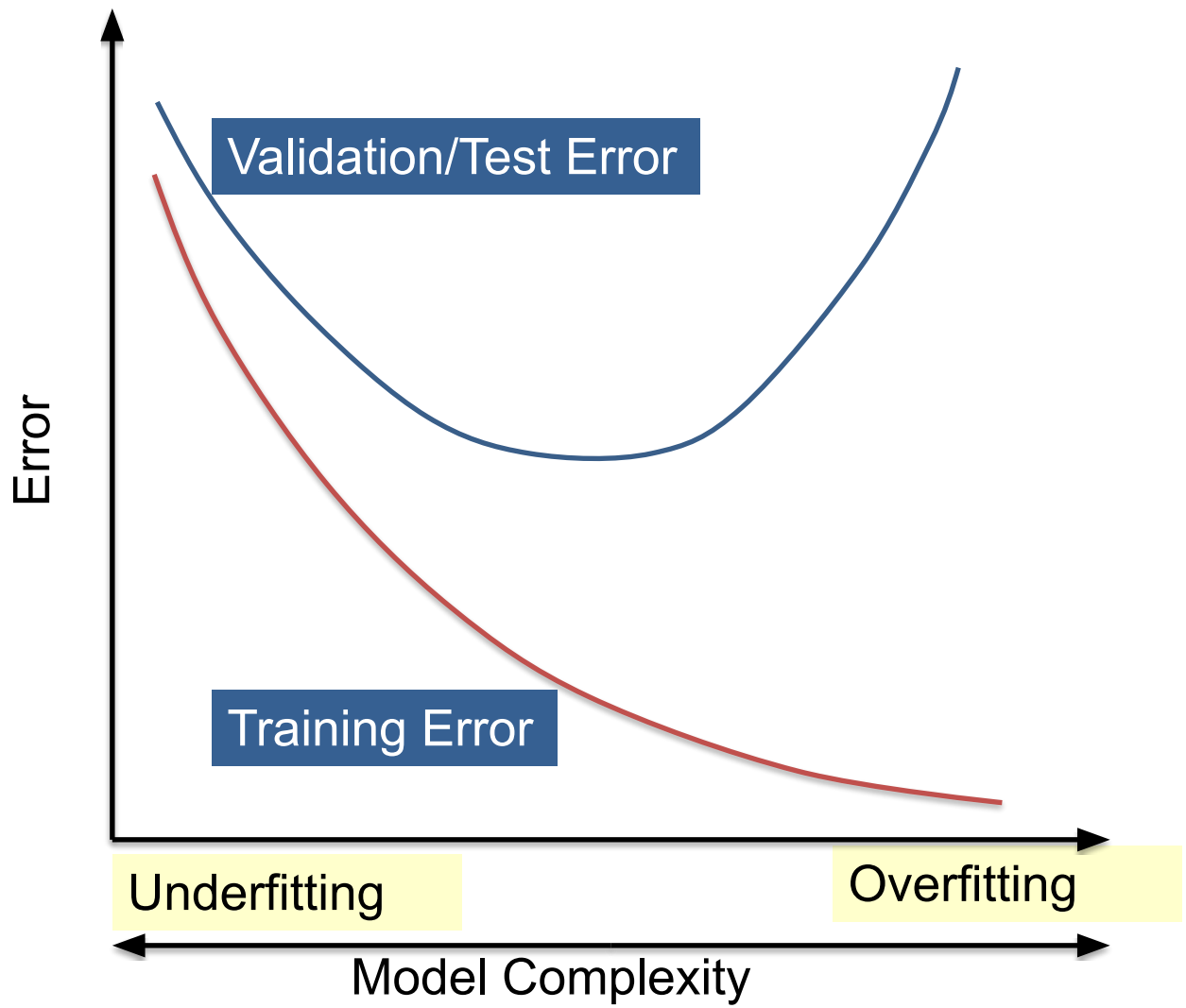
and

- relatively rigid models have high bias and low variance.

Bias variance trade-off

Overall goal : to minimize the expected loss, which can be decomposed into the sum of a (squared) bias, a variance, and a constant noise term.

The model with the optimal predictive capability is the one that leads to the best

balance between bias and variance.

Underfitting vs Overfitting (Bias-Variance Tradeoff)